

RAMRACHAI MARMA

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Skills

Machine Learning & AI: Machine learning, GenAI, Azure OpenAI (GPT-4), NLP, RAG pipelines, prompt engineering, secure LLM deployment, Azure Document Intelligence

Programming & Automation: Python, Python automation, Power Automate, GitHub CI/CD, Azure DevOps CI/CD

Cloud & Infrastructure: Microsoft Azure, Azure AD, Azure DevOps, AWS (basic), Google Cloud Platform (basic)

Data & Analytics: SQL, Snowflake, Power BI, Qlik Sense, data pipelines, data engineering

Methodologies: Agile/Scrum, requirements gathering, process modelling

Security & Governance: IT security engineering, cloud security, access governance, risk assessment, IT audit

Languages: English (C1), German (B1, developing)

Professional Experience

Working Student, IT Security & Cloud Governance

June 2024 – Present

Siemens Mobility GmbH / Erlangen, Germany

- Assessed and documented gaps in access control, logging, and traceability across cloud and industrial IT workflows, supporting risk assessment and security governance objectives.
- Hardened Azure and Azure DevOps CI/CD pipelines by introducing automated security checks, improving deployment visibility across development environments.
- Automated recurring cloud deployment tasks using Python and Power Automate, reducing manual effort and improving operational reliability.
- Operational and incident data was analyzed in Qlik Sense to insight relevant to surface reliability and security, enabling data-driven engineering decisions.
- Communicated technical risk and security findings to non-technical stakeholders across teams, bridging IT security and business goals.

Master's Thesis: AI-Assisted Railway Fault Reporting System

January 2025 – June 2026

Siemens Mobility GmbH / Erlangen, Germany

- Designed and developed an end-to-end AI-powered fault reporting platform that converts unstructured German railway maintenance reports into structured, standardized records using Streamlit, Azure Functions, Azure OpenAI GPT-4o, Azure Blob Storage, and SharePoint.
- Architected a dual AI workflow: AI Analysis extracts structured fault information from free-text maintenance reports using few-shot prompting with historical reports, while AI Categorization maps each report to a controlled 125-category bilingual engineering taxonomy for standardized classification.
- Built an enterprise backend using Azure Functions to orchestrate AI services, integrate Azure OpenAI, manage historical context stored in Azure Blob Storage, and synchronize approved reports with SharePoint.
- Developed a human-in-the-loop review workflow where technicians validate, edit, and approve AI-generated suggestions before reports are permanently stored, improving reliability and reducing manual documentation effort.
- Created an automated document ingestion and preprocessing pipeline (DATA_Extractor) that consolidated 272 PDF, DOCX, and JSON engineering documents into a structured JSONL knowledge base using Azure Document Intelligence for layout-aware extraction.

Data Analyst

May 2021 – May 2022

aiQuest Intelligence / Remote

- Worked with operational and learner data to improve reporting quality, validate datasets, and support business decisions within an online learning platform.
- Data cleaning, trend analysis, and report preparation were performed to help stakeholders monitor platform performance and operational metrics.

Volunteer Instructor

January 2021 – March 2022

Rural Education Initiative / Chittagong, Bangladesh

- Taught mathematics and logical reasoning to students, adapting materials to different skill levels, and tracking progress through regular performance checks.

Finance Data Analyst Intern

August 2017 – December 2017

Oasis Group / India

- Assisted with financial data management, record verification, and reporting processes to improve data accuracy and support internal financial operations.
- Reviewed financial records for completeness and consistency, documented discrepancies, and collaborated with the finance team to maintain reliable reporting.

Selected Projects

- **RAG Chatbot, Secure AI Pipeline:** Built a production-ready RAG pipeline with automated document ingestion, semantic search, and controlled data access, applying security and access governance controls throughout.
- **Requirements Specification Extraction Tool:** Developed an AI tool using Azure Document Intelligence to extract and deduplicate structured requirements from documents in various formats (PDF, Word, Excel, CSV).
- **Data Engineering Pipeline:** Automated an end-to-end data pipeline in Python with structured transformation workflows and GitHub CI/CD, enabling repeatable, auditable processing.

Education

M.Sc. Data Science | Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany | April 2023 – June 2026

B.Eng. Information Technology | Gujarat Technological University, India | June 2016–September 2020

Certificates

- Get Started with Python, Google/Coursera (Certificate ID: UYKMUAN9VFW2)
- Foundations of Data Science, Google/Coursera (Certificate ID: F4ZWYPFMX3CF)
- Foundations: Data, Data, Everywhere, Google/Coursera (Certificate ID: 8BAUVBG8SCTK)
- SQL, MySQL for Data Analytics and Business Intelligence, Udemy (Certificate ID: UC-f644c030-4c86-4c68-b651-bc1e7745bc95)

Awards

- Fully Funded Government Scholarship (ICCR, Government of India): competitive merit-based award for study abroad.